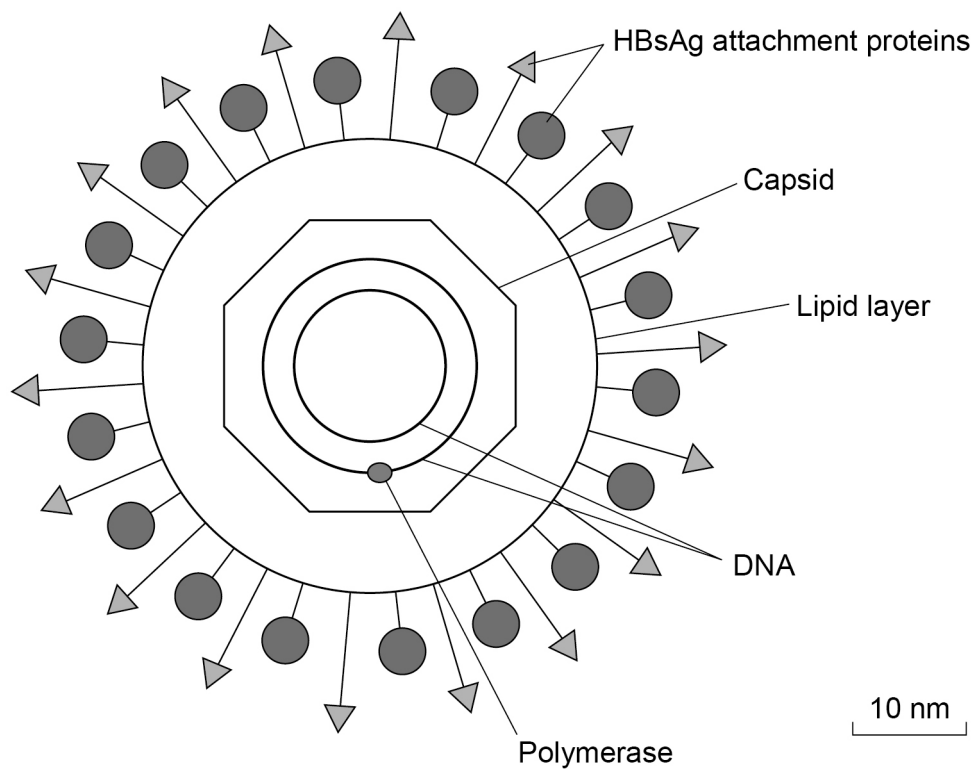


07

Hepatitis B is a life-threatening liver infection caused by the hepatitis B virus (HBV).

Figure 7 shows the structure of HBV.

Figure 7



07

.1

HBV infects a liver cell. The liver cell is $25\mu\text{m}$ in diameter.

Use **Figure 7** to calculate how many times larger in diameter this cell is than HBV. You should use the lipid layer to measure the diameter of HBV.

[2 marks]

Answer _____ times larger

Scientists investigated the effectiveness of two types of RNA interference (RNAi) molecules on reducing HBV replication. These molecules were:

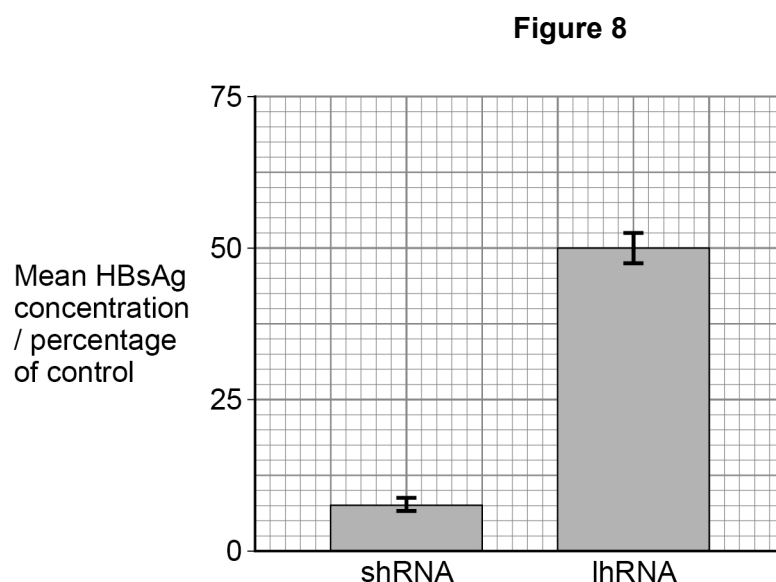
- short hairpin RNA (shRNA)
- long hairpin RNA (lhRNA).

The scientists infected mouse liver cells with HBV and transferred either shRNA or lhRNA into these cells. Then they determined the concentration of the attachment proteins, HBsAg, in these cells.

The concentration of HBsAg is a measure of HBV replication.

Figure 8 shows the scientists' results.

The error bars represent ± 2 standard deviations from the mean, which includes over 95% of the data.



0 7 2

One method of transferring RNAi molecules into cells involves combining these molecules with a lipid. Suggest why this increases uptake of RNAi molecules into cells.

[1 mark]

Question 7 continues on the next page

Using all the information provided, evaluate the use of the two types of RNAi in treating hepatitis B in humans.

Do **not** refer in your answer to how RNAi reduces HBV replication.

[5 marks]

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